$$\omega = \omega_0 \exp\left(-\frac{GM}{c^2 R}\right) = \omega_0 \exp\left(-\frac{v_{II}^2}{2c^2}\right), \tag{25}$$

$$v_{II} = \sqrt{\frac{2GM}{R}}\,. \tag{26}$$

(19) (25)

$$\left(\frac{\Delta\omega}{\omega}\right)_0 = 1 - \exp\left(-\frac{v_{II}^2}{2c^2}\right)\sqrt{\frac{1-v_X/c}{1+v_X/c}}\,. \tag{27}$$

$$v_{II} = \sqrt{2\ln\left(\frac{\sqrt{\frac{1-v_X/c}{1+v_X/c}}}{1-\left(\frac{\Delta\omega}{\omega}\right)_0}\right)}\,c = 2.83\cdot 10^6 \text{ m/s}\,. \tag{28}$$

제 C 부. 움직이는 매질 속의 빛 (4,0점)

3.1 **[1,1점]** S'

$$u_x' = \frac{dx'}{dt'}, \tag{29}$$

$$u_y' = \frac{dy'}{dt'}. \tag{30}$$

S

$$u_x = \frac{dx}{dt}, \tag{31}$$

$$u_y = \frac{dy}{dt}. \tag{32}$$

$$dx = \frac{dx' + V dt'}{\sqrt{1 - V^2/c^2}}, \tag{33}$$

$$dy = dy', \tag{34}$$

$$dt = \frac{dt' + dx' V/c^2}{\sqrt{1 - V^2/c^2}}\,. \tag{35}$$

(33)-(35) (29)-(32)

$$u_x = \frac{u_x' + V}{1 + \frac{u_x' V}{c^2}}, \tag{36}$$

$$u_y = \frac{\sqrt{1 - V^2/c^2}}{1 + \frac{u_x' V}{c^2}}\,u_y'\,. \tag{37}$$

3.2 **[1,4점]** .

(14) , α'